

Lectures	Lecture Topics	Lab
Week 1 8/19 8/21	Introduction and Expectations Analog Circuits • Ohm's law • Circuit diagrams • Circuit components- Resistors, Capacitors, Inductors	Lab 1: Room 142 Safety Lecture. Familiarization with equipment carts - function generator, oscilloscope, power supply, breadboard, multi-meter. $E=IR$ circuit. Power limitation of a resistor.
Week 2 8/26 8/28	Intro to Data Acquisition Analog Circuits (continued) • Voltage divider circuit analysis • RC circuits • DC power supplies • Junction diodes	Lab 2: Room 142 Introduction to NI DAQ device - Write Matlab program and acquire signal from a function generator. Observe function generator signal on oscilloscope. Plot data. Write data to file for post processing. Lab notebook check.
Week 3 9/2 9/4	9/2 LABOR DAY – no lecture 9/4 Lecture Labs will be held this week.	Lab 3: Room 142 Build and measure properties of basic circuits, such as voltage divider, RC filters.
Week 4 9/9 9/11	Analog Circuits (continued) • Op Amp basics • Inverting and non-inverting amplifiers • Difference amplifiers • Other op amp applications	Lab 4: Room 142 OpAmp experiments - Open loop gain of op amp. Gain of inverting and non-inverting amplifier.
Week 5 9/16 9/18	Analog Circuits (continued) • Filtering noisy signals • Concept of frequencies in a signal • Amplifiers and filters • High, low and band pass filters	Lab 5a: Room 142 Amplification, biasing and filtering with op amp circuits.
Week 6 9/23 9/25	Digital Circuits • Digital basics • Digital representations of analog signals • Analog to digital / Digital to analog conversion	Lab 5b: Room 142 Continuation of Lab 5.

Week 7 9/30 10/2	9/30 - Exam 1 Written part (40% of Exam 1 total) 10/2 – Lecture Flowmeters	Exam 1: Room 142 Practical part - circuit building (60% of Exam 1 total) Lab notebook turned in for a grade.
Week 8 10/7 10/9	10/7 FALL BREAK – no lecture 10/9 Lecture Labs will be held this week.	Lab 6: Room 028 Flow rate measurement with a rotameter, venturi, and orifice meters.
Week 9 10/14 10/16	Statistics Velocity and Flow Rate Measurements - Pressure measurement - Static performance of measuring systems Pitot Tubes and Airspeed	Lab 7: Room 142 Pressure and velocity measurements with a pitot-static tube. Processing data with Matlab.
Week 10 10/21 10/23	Statistics Velocity and Flow Rate Measurements Pitot Tubes and Airspeed (Continued)	Lab 8: Room 142 Conservation of mass and momentum in an axisymmetric jet. Lab report-1 on this lab due at lab time week of November 4.
Week 11 10/28 10/30	Surface Pressure Measurements: Lift and Drag on an Airfoil	Lab 9: Room 142 Measuring lift and drag on a wing using surface pressures. Lab report-2 on this lab due at lab time week of November 18.
Week 12 11/4 11/6	Aerodynamic Heating and Heat Transfer by Conduction, Convection and Radiation - Steady one dimensional conduction - Convective heat transfer to/from flat plate - Newton's law of cooling	Lab 10: Room 142 Heated cylinder in cross flow – Determine heat transfer coefficient. Determine local and average Nu, Lab notebook check. Lab report 1 due at lab time. Lab report-3 on this lab due on Monday November 25.
Week 13 11/11 11/13	Measurements in Supersonic Flow - Schlieren Visualization - Pitot tube, velocity measurements	Lab 11a: Room 142 Set up pressure sensors and Schlieren visualization system to measure Mach number in supersonic wind tunnel.
Week 14 11/18 11/20	Measurements in Supersonic Flow (continued)	Lab 11b: Room 142 Supersonic wind tunnel experiment to observe and photograph oblique shock waves. Lab report 2 due at lab time.
Week 15 11/25 11/27	Measurements in Supersonic Flow (continued) 11/25 Lecture 11/27 THANKSGIVING BREAK – no lecture	No labs due to Thanksgiving Break. Lab report-3 due on Monday November 25.
Final Exam Week 12/2	Exam 2 - Location and Time to be Announced	No labs. Lab notebook turned in for a grade.

Also, by appointment.

Required Textbook: "Hands-On Electronics, A Practical Introduction to Analog and Digital Circuits", Fourth Edition, by Daniel M. Kaplan and Christopher G. White, Cambridge University Press, ISBN 978-0-521-85391-0.

Grades:

- Based on a curve.
- Lab reports – 30%
- Homework – 10%
- Attendance and participation in labs – 10%
- Quality of laboratory notebook - 10% (5% midterm review + 5% final review)
- Exam 1– 20% (12% practical 8% written)
- Exam 2 – 20% (written only)

Labs:

Participation in labs counts toward 10% of the grade. Just like the lecture, labs will start on time and if the student arrives late, it will be the responsibility of the student to catch up.

If the student misses their regularly scheduled lab session, then the student must make arrangements with the Instructor and Teaching Assistant to complete the lab that week. If the lab is not completed that week, the student cannot turn in a lab report (where applicable) and in addition the final grade will be reduced by one letter grade for each lab missed.

If the student cannot attend a lab for the entire week due to a medical reason or other approved absence, it is the responsibility of the student to arrange a review of the lab with the Instructor or Teaching Assistant. After the review, the student must obtain the lab data from one of their lab partners and write up the lab in their lab notebook as required. The student will complete these activities before the date the lab notebook is required to be turned in for grading.

The grade for lab reports and notebooks turned in late will be reduced by 20%. Late lab reports will no longer be accepted for grading one week after the due date or after the second exam has been given, whichever is earlier. Late lab notebooks will no longer be accepted for grading one week after the first exam has been completed. Lab notebooks must be turned in with the second exam. No exceptions.

Each student will have access to an equipment cart for use during the semester. Students are responsible for the care and well-being of the equipment on the cart. It will be the responsibility of the student to notify the instructor or teaching assistant of any missing or damaged equipment from the cart. Carts will be self-assigned in groups of two (minimum) or four (maximum).

Lab notebooks will be provided to the students. Each student must keep a detailed lab notebook containing all information collected during the labs, which will be turned in at the end of Exam 1 and 2.

For each lab, the student will need to record:

- Date of the experiment
- Name of experiment and Goal of experiment (Use multiple sections if the lab involves multiple parts)
- List of teammates
- Experimental setup
 - Equipment used for the work (e.g. BK Precision 2704c multimeter...)
 - Relevant, data acquisition parameters (number of samples, acquisition rate, etc...)
 - Sketch of experiment with clear labels; photographs are acceptable
 - Procedure including Matlab script (where applicable)
- Summary of results
 - Table of measurements / graphs with titles
 - Brief discussion and conclusions
- Any piece of information (e.g. printed graphs, tables) in your lab notebook must be taped.
- Answers to any questions that are listed in the lab handouts.

The lab notebook will not be as detailed as a lab report but needs to contain sufficient information for someone else to closely reproduce your experiments.

The student is allowed to use the lab notebook for Exam 1 (written and practical parts) and Exam 2. Fill out the lab notebook as you go during lab time. Don't wait until later to start writing your notes, you won't remember what the equipment was, or what parameters you used.

Three lab reports are due throughout the semester. The lab report format and further information will be given at the appropriate time.

Exams:

No make-up exams will be given except for emergency situations. Contact the [REDACTED] and lab reports (where applicable).

Homework:

Homework will be assigned at various times throughout the semester. Each student must turn in a unique homework assignment on paper. Electronic versions of the homework will not be accepted. Tablet printouts will be accepted.

Homework assignments will be due usually at lab time. Homework assignments turned in after the lab session will be considered late. The score for a late homework assignment will be reduced by 20%. Homework assignments will no longer be accepted one week past the due date.

ADA Statement:

Reasonable accommodations will be made for students with documented disabilities. In order to receive accommodations, students must obtain a letter of accommodation from the Center for Disability Resources. The Center for

located at <https://www.iit.edu/cdr/services/reasonable-accommodations>.

Academic Honesty and Collaboration Policy:

IIT expects students to maintain high standards of academic integrity. Students preparing for the practice of a profession are expected to conform to a code of integrity and ethical standards commensurate with the high expectations society places on practitioners of a learned profession. No student may seek to gain an unfair advantage over another. The Code of Academic Honesty is explained in the IIT Student Handbook, and all students are expected to know and adhere to this code.

Departmental policies on academic honesty are consistent with those of the University. The Department policies are provided at https://www.iit.edu/sites/default/files/2021-01/honesty_policy_-_mmae.pdf.

Collaboration on homework is not allowed.

Collaboration on the laboratory notebooks and laboratory reports is not allowed. Each student should write their own lab reports, and, on their own, analyze and plot the data acquired in lab. Plagiarism, which is "an act or instance of using or closely imitating the language and thoughts of another author without authorization and the representation of that author's work as one's own, as by not crediting the original author" is not allowed. Definition from dictionary.com.

Collaboration, plagiarism or the use of artificial intelligence (AI) on a lab report will result in a zero grade for the lab report. A repeated violation will result in the lowering of the final grade by one letter grade.

Collaboration on Exam 1 and 2 is not allowed.

Infractions of the Honesty and Collaboration policy will be referred to the Academic Honesty Committee.

Classroom Expectations Regarding Diversity, Equity, and Inclusion:

The MMAE Department statement on Diversity, Equity, and Inclusion including the incident reporting process is found at <https://www.iit.edu/sites/default/files/2021-01/MMAE-Statement-On-Diversity-Equity-and-Inclusion.pdf>.

It is expected everyone will treat each other with empathy and civility both in person and online.